

MSCI151: Group Methex (Lent term)

- You should work on this assignment in groups of 4 students.
- You will give a presentation during the week beginning March 6, 2023 (Week 18). Your presentation slides should be uploaded to Moodle by **2:00pm on March 6, 2023**.
- Please name your presentation with your group name (e.g. “GroupC.pptx”, where C is your group name).
- Only one member of your group needs to upload the presentation slides.
- The maximum length of your presentation should be 15 minutes. You should also be prepared to answer questions at the end of your presentation, which might last up to 5 minutes.
- It is not necessary for all members of your group to speak during the presentation, but all members should attend and be prepared to answer questions.

Part A

A local grocery store sells two types of energy drinks: “Sprint” and “Vitality”. The store has asked for your help to work out stock control policies for both drinks. Daily demands for the drinks can be unpredictable, and the store wants to work out (i) when they should place orders for new stock, and (ii) how many units (bottles) of stock they should order at a time. Their main objective is to minimise average daily inventory costs – taking into account holding costs, ordering costs and shortage costs for both products.

The store has provided you with the following information about costs for the two drinks:

	Sprint	Vitality
Holding cost per unit per day	£0.05	£0.06
Shortage cost per unit per day	£1.20	£1.80
Fixed ordering cost	£30.00	£30.00
Ordering cost per unit	£0.60	£0.80

The store’s inventory control policy works as follows: deliveries of new stock are received at the beginning of the day. After receiving new stock, the store then incurs holding costs based on the current stock levels of both drinks. During the day, demands are received for both drinks. If the amount of demand for a drink on a particular day exceeds the stock level at the beginning of that day, then the store incurs shortage costs due to the loss of sales. At the end of the day, the store inspects the stock levels for both drinks and decides whether or not to place orders for new stock. They wish to use the following type of system:

- If the current stock level for Sprint, plus the number of units of Sprint that they have ordered but are still awaiting delivery of, is less than or equal to a certain fixed number (call this L_S), they will place an order for a certain fixed number of units of Sprint (call this Q_S).
- If the current stock level for Vitality, plus the number of units of Vitality that they have ordered but are still awaiting delivery of, is less than or equal to a certain fixed number (call this L_V), they will place an order for a certain fixed number of units of Vitality (call this Q_V).

The numbers L_S and L_V are known as “re-order levels”, and the numbers Q_S and Q_V are known as “order quantities”. Note that on a particular day the store may decide to place orders for both drinks, or only one type of drink (e.g. Sprint only), or neither of the drinks.

The two drinks are ordered from different suppliers – call these Supplier A (for Sprint) and Supplier B (for Vitality). The lead time for Supplier A is expected to be 4 days. This means that when an order is placed, it will usually arrive 4 days later. However, based on past experience, the store believes there is a 10% chance that Supplier A will deliver one day late (i.e. after 5 days). For Supplier B, the lead time is expected to be 3 days, but there is a 15% chance that they will deliver one day late (i.e. after 4 days). If an order is placed at the end of Day n , then it will be received at the beginning of Day $n + t + 1$, where t is the lead time. For example, if an order is placed at the end of Day 10 and the lead time is 4 days, it will be received at the beginning of Day 15.

As an example (to show how the process works in more detail), suppose that at the beginning of a particular day, the store has 50 bottles of Sprint in stock, and they are also waiting for 70 bottles to be delivered – but these have not yet arrived. Also suppose they are using a re-order level $L_S = 100$ and an order quantity $Q_S = 70$. Then:

- The holding costs incurred are $£(50 \times 0.05) = £2.50$.
- The shortage costs depend on the daily demand. If the daily demand is 50 or less, there will be no shortage (the store can satisfy all of the demand). But if the demand is 60 (for example), then there will be a shortage of $(60 - 50) = 10$, so the shortage cost will be $£(10 \times 1.20) = £12$.
- Whether or not the store places an order at the end of the day also depends on the demand. They are already waiting for an order of 70 units to be delivered, so if the stock level at the end of the day is more than 30 (i.e. if the daily demand is less than 20), then their stock level (including units being delivered) will be more than 100, which is the value of L_S . So in that case they will not place an order. But if the daily demand is 30 (for example), then the stock level (including units being delivered) will be $20 + 70 = 90$, and this is smaller than L_S , so they will place an order for $Q_S = 70$ units to be delivered. The cost of this order is given by adding the fixed order cost to the number of units ordered times the ordering cost per unit, so it will be $£(30 + 70 \times 0.60) = £72$.

The total inventory costs for the day are given by adding up the holding costs, shortage costs and ordering costs described above. This must be done for both drinks.

To help you understand the demand patterns for the two drinks, the store has provided you with a dataset showing a history of daily demands for the two drinks over a period of one year (365 days). In some cases these demands have been estimated, because they do not know exactly how many bottles they would have sold on days when they were unable to satisfy all demand. See the file **demand_history.xlsx** for the demand history. As you can see, the demands vary significantly from day to day, so you should try to find suitable random distributions to model the variations in demand for both drinks.

The store has asked you to help them set the re-order levels (L_S and L_V) and the order quantities (Q_S and Q_V) in order to minimise their average daily costs. In order to do this, you should set up a simulation model (using Excel or other software of your choice) and experiment with changing the values of L_S , L_V , Q_S and Q_V in order to find the values that give the smallest possible average daily costs.

Can you find the optimal values of L_S , L_V , Q_S and Q_V , and what is the average daily cost if your recommended values are used?

(Please turn over.)

Part B

The store would also like you to investigate another question. They have the option of ordering the “Sprint” drink from a different supplier – call this Supplier C. Supplier C offers to deliver orders more quickly than Supplier A, so if Supplier C is used, the lead time will usually be only 2 days, but there is also a 10% chance that the lead time will be 3 days. However, Supplier C charges a fixed ordering cost of £40, so it is more expensive to place an order with Supplier C compared to Supplier A. The store wants you to investigate whether they should order Sprint from Supplier A or Supplier C. (Note that if Supplier C is used, the values of L_S and Q_S could be different from the values used for Supplier A. In other words, the store is able to make adjustments to the re-order level and order quantity depending on which supplier is used.)

Do you recommend that the store should order Sprint from Supplier C, or should they continue to use Supplier A? If Supplier C is used, what should the re-order level and order quantity be?

Part C

Now suppose that Suppliers A and B have merged together, so that there is now a single supplier that is able to supply both drinks (forget about Supplier C in this part). This means that the store can save money on the fixed ordering costs by placing orders for both drinks at the same time. This works as follows: if the store places orders for both Sprint and Vitality on the same day, then they only need to pay the fixed ordering cost of £30 once. For example, if they order 70 units of Sprint and 80 units of Vitality on the same day, the total cost of the order will be $£(30 + 70 \times 0.60 + 80 \times 0.80) = £136$. If they place an order for only one of the drinks on a particular day, the fixed ordering cost will still be the same as before, i.e. £30.

In this part you may assume that if an order is placed (whether it’s for both drinks or only one of the drinks), the lead time will usually be 4 days but there is a 10% chance that it will be 5 days.

Given this new situation, the store would like to ask you what their ordering policy should be in order to take advantage of the possibility of saving money by placing orders for both drinks at the same time. In this part, you can experiment with different types of ordering systems that work differently from the ones in Part A and Part B. For example, you might try using a rule where the re-order level for one drink on a particular day becomes smaller if an order for the other drink has already been placed on the same day (to make it more likely that the store will place orders for both drinks at the same time).

What should the store’s ordering policy be in order to minimise average daily costs?

(Please turn over.)

What you need to do (and some further advice!)

Your task is to prepare a presentation (maximum 15 minutes) to advise the store on their ordering policies in the situations described in Parts A, B and C.

It is expected that you will use Excel (or other software of your choice) to construct simulation models in order to answer these questions. You are not expected to do any simulation ‘by hand’ – it is a computer-based exercise. You will need to submit your final presentation slides (e.g. PowerPoint files) by 14:00 on March 6, 2023. Your Excel files do not need to be submitted.

Note that although the store is mainly interested in minimising average daily costs, they are also interested in other performance measures – for example, the frequency of “stockouts” and the percentage of demand they are able to satisfy under your recommended ordering policies. Therefore it would also be a good idea to include information about other performance measures in your presentation.

Your presentation should include:

- **A brief introduction, including an overview of your simulation model(s);**
- **Some details of how you set up the simulation model(s), including the probability distributions used to model uncertain variables and how you decided upon these;**
- **An explanation of the different ordering policies that you considered and how you attempted to find the ‘best’ ordering policy in each part;**
- **Some discussion of other performance measures, e.g. percentage of demand satisfied (see above);**
- **Conclusions and recommendations for the company.**

Credit will also be given for using well-designed presentation slides. You should try to use figures and tables in order to explain your findings (not just text). You are also advised to review the “Presentation Skills” resources on Moodle and try to avoid common mistakes, such as making your slides too text-heavy.

You are reminded of the importance of performing multiple replications of your simulation. Also, try to take into account other issues that have been discussed in the simulation lectures. It’s a good idea to look over all of the lecture slides again.

After you have created your simulation model, try to do a “sense check”. Do the results of the simulation seem fairly reasonable, or are you getting “crazy” results – e.g. average daily costs that seem much higher or lower than expected? If the results seem crazy, then you may have made a mistake somewhere in the logic of the simulation model (or maybe your ordering policy just doesn’t make sense).

The marking sheet that will be used to assess your presentation is available on Moodle – see <https://modules.lancaster.ac.uk/mod/resource/view.php?id=1946739>. You may find it useful to look at this to see how your presentation will be assessed.

If you have any other questions related to this assignment, please post them in the ‘Questions and Answers’ forum on Moodle – see <https://modules.lancaster.ac.uk/mod/forum/view.php?id=1946727>.